

Question	Answer	Marks
4(a)(i)	thermodynamic temperature	B1
4(a)(ii)	molar gas constant	B1
4(b)(i)	m : mass of one molecule (of the gas)	B1
	$\langle c^2 \rangle$: mean-square speed (of molecules)	B1
4(b)(ii)	$NBT / A = \frac{1}{3} Nm \langle c^2 \rangle$	M1
	clear use of $E_K = \frac{1}{2} m \langle c^2 \rangle$ leading to $E_K = 3BT / 2A$	A1
4(c)	line with positive gradient passing through the origin	B1
	smooth curve with decreasing positive gradient	B1

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